



GREEN SKILLING, SUSTAINABILITY & AI

Building an Eco-Conscious Future Through Technology and Collaboration

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About Me



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SEMESTER

5th Semester

A Short Introduction

I am a third-year Computer Science Engineering student passionate about applying machine learning to real-world problems. My interest in sustainability and AI grew from exploring how intelligent systems can monitor pollution, optimise energy use and support the UN Sustainable Development Goals — motivating me to combine my technical skills with environmentally responsible innovation.

Things I Learned

Key takeaways from the learning resources explored during this module



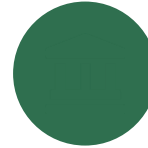
SDG 17

Partnerships for the Goals — global collaboration to achieve sustainable development, especially reducing inequality.



Green Data Centres

How data infrastructure is shifting toward renewable power and energy-efficient cooling systems.



Government Initiatives

National missions like Digital India and Smart Cities driving sustainability at scale.



Renewable Energy

Solar, wind and clean power sources reducing dependence on fossil fuels.



AI for Environment

Machine learning applications in monitoring pollution, climate and natural resources.

SDG 17: Reduce Inequality Through Partnership

17

of 17 UN Sustainable
Development Goals

SDG 17 focuses on strengthening global partnerships — between governments, businesses, communities and individuals — to mobilise the knowledge, technology and finance needed to reduce inequality and achieve sustainable development for everyone.



Why Collaboration Matters

No single country or organisation can solve climate change or inequality alone — shared resources and knowledge accelerate progress.



How AI Helps Achieve SDG 17

AI enables data-sharing platforms, predictive analytics and open-source tools that connect governments, researchers and communities worldwide.



Global Impact

Cross-border cooperation on technology transfer and funding helps developing regions leapfrog straight to clean, inclusive growth.

Learning from Data Centre Documentaries

Understanding the environmental footprint of the digital world



Why Data Centres Consume So Much Energy

Thousands of servers run 24/7 for computation, storage and cooling — making data centres among the largest industrial power consumers globally.



Green Data Centres

Facilities are being redesigned with energy-efficient hardware, smart layouts and sustainable materials to cut waste.



Renewable Energy Usage

Companies like Google and Microsoft now power data centres with solar and wind energy to reduce fossil-fuel dependence.



AI-Based Cooling & Power Optimisation

Machine learning models predict cooling needs in real time, cutting energy use — Google reported major cooling energy savings using DeepMind AI.



Carbon Emission Reduction

Efficient design plus renewable power together significantly lowers the carbon footprint of global cloud infrastructure.

Learning from Government Websites

Exploring national platforms driving India's sustainability mission



Digital India

A national initiative to make government services digital, transparent and accessible — reducing paper use and improving efficiency.



Smart Cities Mission

Building urban infrastructure with efficient energy grids, waste management and clean mobility.



Renewable Energy Schemes

Government subsidies and missions promoting solar, wind and clean-energy adoption nationwide.



E-AMRIT Portal

A gov platform helping citizens compare EV vs petrol vehicle emissions and calculate CO2 savings.



Skill Council for Green Jobs (SCGJ)

A government body creating job-ready certifications for the renewable energy and green economy workforce.



Environmental Conservation Programs

National forestry, afforestation and pollution-control missions protecting natural ecosystems.

Activity Screenshots



Data center map



What is Green Skilling?



Definition

Green skilling is the process of gaining knowledge and technical abilities needed for jobs that support environmental sustainability — from renewable energy to eco-friendly manufacturing.

Why It Matters

As industries shift toward net-zero goals, demand for a green workforce is rising fast — making these skills essential for future employability.

Career Opportunities

Solar technician, EV engineer, sustainability analyst, environmental data scientist, and green-AI developer are among the fastest-growing roles.

Examples in Practice



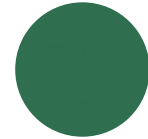
Renewable Energy



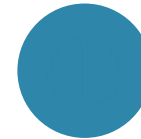
Electric Vehicles



Waste Management



Sustainable Agriculture



AI-Based Environmental Solutions

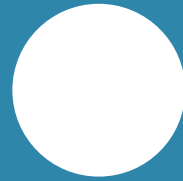
What is Sustainability?

Sustainability means meeting the needs of the present without compromising the ability of future generations to meet their own needs.



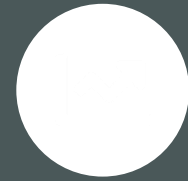
Environmental

Protecting ecosystems, biodiversity, air, water and natural resources for the long term.



Social

Ensuring equity, health, education and wellbeing for communities everywhere.



Economic

Building growth that is efficient, inclusive and doesn't deplete future resources.

Everyday Sustainable Practices

- ✓ Save electricity & water at home
- ✓ Choose public transport or EVs
- ✓ Reduce, reuse and recycle waste
- ✓ Support local & sustainable products

AI — Benefits and Challenges for Sustainability



Benefits



Smart Agriculture

AI-guided irrigation and crop monitoring



Climate Prediction

Forecasting extreme weather and climate shifts



Waste Segregation

AI-smart bins that sort recyclables automatically



Energy Optimisation

Smart grids balancing supply and demand



Water Management

Detecting leaks and optimising usage



Forest Monitoring

Satellite AI tracking deforestation in real time



Challenges



High Energy Consumption

Training large AI models requires significant power



Data Centre Emissions

Cloud infrastructure adds to global carbon output



Electronic Waste

Rapid hardware upgrades increase e-waste volumes



Water Usage for Cooling

Large data centres consume vast amounts of water



Need for Responsible AI

Ethical, transparent and efficient AI design is essential



EcoVision AI — Intelligent Environmental Monitoring System



Problem Statement

Environmental degradation and pollution often go undetected until damage is severe, due to a lack of real-time, accessible monitoring tools for communities and authorities.



Objectives

- ✓ Build a real-time environmental monitoring platform
- ✓ Detect air quality and pollution levels using AI
- ✓ Predict pollution trends to enable early action
- ✓ Encourage eco-friendly behaviour via a live dashboard



Technologies

Python

Machine Learning

Computer Vision

IoT

Cloud



Key Features



Air quality monitoring



AI waste classification



Pollution prediction models



Carbon footprint dashboard



AI chatbot assistant



Expected Impact: Supports SDG 3, 11, 13 & 17 by enabling data-driven, community-focused environmental action.



Thank You